

Peng et al. (2023) conducted the first controlled experiment to measure the impact of GitHub Copilot on productivity in professional software development. The authors recruited 95 programmers through Upwork and randomly assigned them to treatment and control groups. Participants were tasked with implementing an HTTP server in JavaScript. The group with access to Copilot completed the task 55.8% faster than the control group. The effects were heterogeneous, with greater benefits observed among less experienced developers and those who spent more hours programming daily. The study highlights the potential of AI tools to support career transitions into software development, although it does not assess the quality of the generated code.

Ziegler et al. (2024) investigated whether usage measurements derived from interactions with GitHub Copilot could predict developers' perceived productivity. The study analyzed 2,047 survey responses paired with telemetry data, using the SPACE framework as a conceptual foundation. The results showed that suggestion acceptance rate was the strongest predictor of perceived productivity, outperforming persistence-based metrics. Junior developers reported the largest productivity gains and exhibited higher suggestion acceptance rates. The authors argue that Copilot's value lies not only in reducing typing effort but also in helping developers make progress toward their goals.

Bird et al. (2023) examined early user experiences with GitHub Copilot through three complementary studies: a forum analysis, a think-aloud study, and a survey. Their findings revealed that developers spent less time consulting Stack Overflow but often had a weaker understanding of how the generated code worked. Participants also reported spending more time reading and reviewing code than writing it. The study discusses shifts in the developer's role and highlights concerns regarding trust in AI-assisted development tools. The authors emphasize the importance of tracking the provenance of AI-generated code.

Wivestad and Ulfsnes (2025) explored perceived productivity changes following the introduction of GitHub Copilot in two agile organizations using the SPACE framework. Unlike previous studies, they measured perceived change rather than agreement with predefined statements, aiming to reduce the risk of overestimating positive effects. Their results identified several perceived benefits, although the overall impact was more conservative than reported in earlier research. Productivity improvements were associated with only four of the nine investigated features, including job satisfaction and workflow efficiency. The authors also identified discrepancies within the SPACE dimensions of "Performance" and "Communication and Collaboration."

Chen et al. (2026) evaluated the validity of different approaches to measuring AI-assisted productivity through a mixed-methods study conducted at BNY Mellon. The research combined 2,989 survey responses with 11 semi-structured interviews. Although satisfaction with GitHub Copilot was generally high, reported time savings were modest, and the correlation between these two measures was relatively weak. The study identified six productivity-related factors, including self-sufficiency, cognitive load, and ease of peer review. It also highlighted long-term dimensions, such as technical expertise development and ownership of work, which have

received limited attention in prior research. The authors advocate for multidimensional and context-sensitive approaches to productivity evaluation.

Coelho et al. (2025) investigated software developers' perceptions of productivity within a Brazilian research and technology organization through two focus groups involving 16 participants, followed by a confirmatory questionnaire. The study identified five productivity factors and four unproductivity factors, each associated with distinct indicators for coding, code review, and documentation activities. Regarding Generative AI, participants reported expected benefits as well as concerns related to hallucinations, code duplication, and the potential reduction of critical thinking among novice developers. The findings were mapped onto the dimensions of the SPACE framework. The authors conclude that productivity is a multidimensional construct that requires balancing efficiency gains with the preservation of technical skills.

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